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Author(s): Richard Darst, Judith Palagallo and Thomas Price

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Fractal Tilings in the Plane

RICHARD DARST
Colorado State University
Fort Collins, CO 80523

JUDITH PALAGALLO*
THOMAS PRICE*
University of Akron
Akron, OH 44325

Introduction

Tilings have appeared in human activity since prehistoric times. They are used in the design of floor and wall coverings for cathedrals, commercial buildings, and personal dwellings. Mathematicians study the geometric structure of tilings. A checkerboard is an elementary example of a *similarity tiling*, one that is composed of smaller tiles (*rep tiles*) of the same size, each having the same shape as the whole. Each rep tile in the checkerboard is the scaled and translated image of the entire board. For the checkerboard in FIGURE 1a, the lower left tile is the image of the checkerboard under

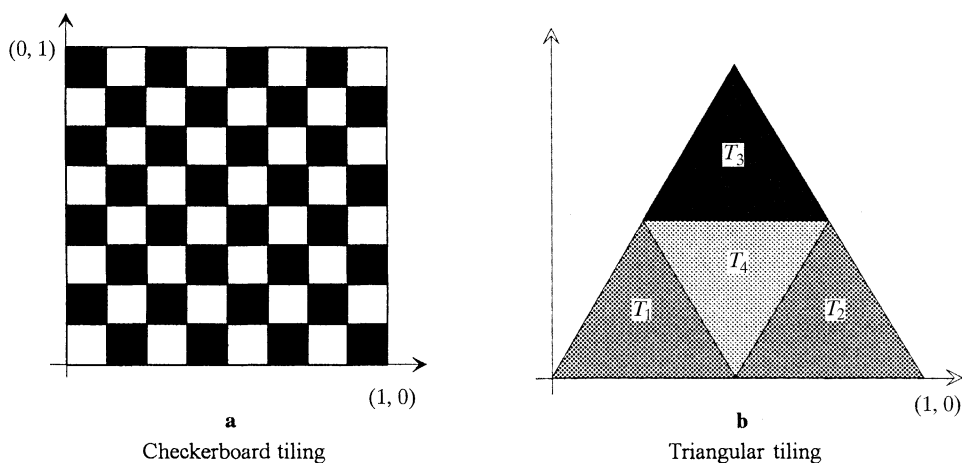


FIGURE 1

the mapping

$$f := \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \mapsto \begin{bmatrix} \frac{1}{8} & 0 \\ 0 & \frac{1}{8} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix},$$

where $\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$ denotes any point on the checkerboard. The whole checkerboard can be formed using translates of this smaller image. One goal of this paper is to investigate the properties of linear mappings that generate similarity tilings.

* Research was conducted while on professional leave from the University of Akron.

In a more general setting the mappings may also involve rotations. For instance, the equilateral triangle T in FIGURE 1b is a similarity tiling since it is composed of four smaller equilateral triangles T_1, T_2, T_3, T_4 . Consider the mappings defined by

$$\begin{aligned} f_1 &:= \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \rightarrow \begin{bmatrix} 0 \\ 0 \end{bmatrix} + \begin{bmatrix} .5 & 0 \\ 0 & .5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \\ f_2 &:= \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \rightarrow \begin{bmatrix} 0.5 \\ 0 \end{bmatrix} + \begin{bmatrix} .5 & 0 \\ 0 & .5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \\ f_3 &:= \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \rightarrow \begin{bmatrix} 0.25 \\ \sqrt{3}/4 \end{bmatrix} + \begin{bmatrix} .5 & 0 \\ 0 & .5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \\ f_4 &:= \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \rightarrow \begin{bmatrix} 0.75 \\ \sqrt{3}/4 \end{bmatrix} + \begin{bmatrix} -.5 & 0 \\ 0 & -.5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}. \end{aligned}$$

Then $T_j = f_j(T)$, $j = 1, \dots, 4$. Notice that the matrices “shrink” the tiling T to a rep tile, and the last matrix also performs a rotation. Adding the fixed vectors to the corresponding rep tiles translates them to their appropriate locations.

The checkerboard and triangle tiles have straight edges. In this paper we are interested in generating tilings with tiles (*fractiles*) whose boundaries are fractal curves. (Various definitions of fractal curves are given in [2], [11], and [12]. However, as Barnsley states in [2, p. 33], fractals are best explained by the many pictures and contexts that refer to them.) We will use an iterative process, involving repeated compositions of two or more functions, to generate these fractal tilings. The functions are constructed from translates of the inverse of a linear transformation $g(z) = Mz$, where M is an invertible 2×2 matrix with integer entries. (Geometric properties used later in this article require that M be an integer matrix.) The inverse transformation $g^{-1}(z)$ is the “shrinking function” that maps the entire fractal tiling onto a fractile. Before outlining the underlying mathematics, we briefly describe the basic algorithm and illustrate it with some examples.

Examples of fractal tilings

As a simple example, consider the matrix $M = \begin{bmatrix} a & -b \\ b & a \end{bmatrix}$, where the integers a and b are chosen so that $a^2 + b^2 > 1$. If we interpret $\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$ and $\begin{bmatrix} a \\ b \end{bmatrix}$ as points in the complex plane, then $M \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} ax_1 - bx_2 \\ ax_2 + bx_1 \end{bmatrix}$ represents complex multiplication of $x_1 + ix_2$ by $a + ib$. Next, we choose a collection of vectors to translate copies of the fractile so that they are positioned correctly in the tiling. Notice that the unit square, determined by the vectors $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$ and $\begin{bmatrix} 0 \\ 1 \end{bmatrix}$, is mapped by M onto the square S , of area $m = a^2 + b^2$, spanned by vectors $\mathbf{v}_1 = \begin{bmatrix} a \\ b \end{bmatrix}$ and $\mathbf{v}_2 = \begin{bmatrix} -b \\ a \end{bmatrix}$. Define the set $\mathcal{E} = \{\mathbf{r}_j\}$ of vectors with integer coordinates that lie in or on S but not on the two outer edges that do not have the origin as a vertex. Then $\mathcal{E}$ contains exactly m vectors; we will use them for the translation vectors $\{\mathbf{r}_j\}$. The tiling can be drawn by a computer using the iterative procedure illustrated in Examples A and B.

Example A. Let $M = \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}$; then $m = 2$. From the square in FIGURE 2a we determine the two translation vectors $\mathbf{r}_1 = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$ and $\mathbf{r}_2 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$. Then $\mathcal{E} = \{\mathbf{r}_1, \mathbf{r}_2\}$, and for $\mathbf{z} = (x_1, x_2)$, we define the mappings $f_j(\mathbf{z}) := \mathbf{r}_j + M^{-1}(\mathbf{z})$ for $j = 1, 2$. That is,

$$f_1 := \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \mapsto \begin{bmatrix} 0 \\ 0 \end{bmatrix} + \begin{bmatrix} .5 & -.5 \\ .5 & .5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$f_2 := \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \mapsto \begin{bmatrix} 1 \\ 0 \end{bmatrix} + \begin{bmatrix} .5 & -.5 \\ .5 & .5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}.$$

To initiate the iteration process we randomly choose any point $\mathbf{z}_0$ in the plane and evaluate $f_1(\mathbf{z}_0)$ and $f_2(\mathbf{z}_0)$. Then for $n \geq 1$, we choose recursively and randomly $\mathbf{z}_n \in \{f_1(\mathbf{z}_{n-1}), f_2(\mathbf{z}_{n-1})\}$. After a few iterations the generated points lie near the tiling. So for $n \geq 100$, plot the points as they are generated. FIGURE 2b shows the result of several thousand iterations. (Increasing the number of iterations may improve the quality of the computed image. Special purpose software for drawing fractal tilings is described at the end of this paper.) The boundary of the spiral, the snowflake curve, is an example of a fractal curve. Mandelbrot [12] showed that the distance along the boundary between any two points is infinite.

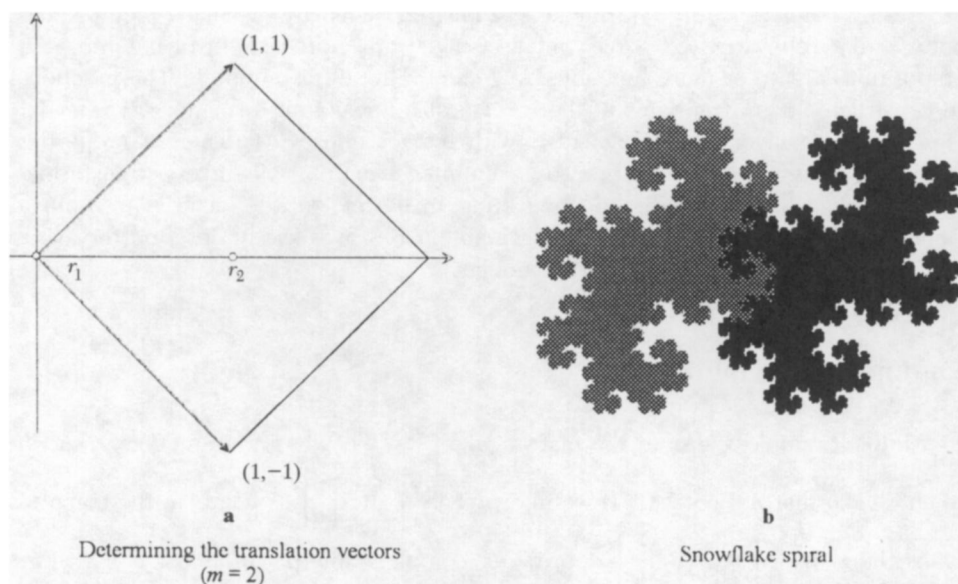


FIGURE 2

The collection of functions $\{f_j\}$ is called an *iterated function system*. The set A of randomly generated points that results from this process is called the *attractor*. Notice that $A = A_1 \cup A_2$, where $A_j = f_j(A)$ and the A_j 's have disjoint interiors.

Using M from Example A, the reader can explore the four different tilings generated when $\mathbf{r}_2$ is replaced by any of $\begin{bmatrix} -1 \\ 0 \end{bmatrix}$, $\begin{bmatrix} 0 \\ -1 \end{bmatrix}$, $\begin{bmatrix} 2 \\ 1 \end{bmatrix}$, and $\begin{bmatrix} 1 \\ 2 \end{bmatrix}$. Observe that the

shape of the tiling may change with the choice of translation vector.

Example B. Let $M = \begin{bmatrix} 1 & 2 \\ -1 & 1 \end{bmatrix}$. As shown in the parallelogram in FIGURE 3a, let $\mathbf{r}_1 = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$, $\mathbf{r}_2 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ and $\mathbf{r}_3 = \begin{bmatrix} 2 \\ 0 \end{bmatrix}$. This choice of vectors produces a tiling with the three tiles stacked horizontally as shown in FIGURE 3b. As an exercise, we recommend that the reader choose translation vectors and generate a tiling for the matrix $M = \begin{bmatrix} 1 & 1 \\ -1 & 2 \end{bmatrix}$.

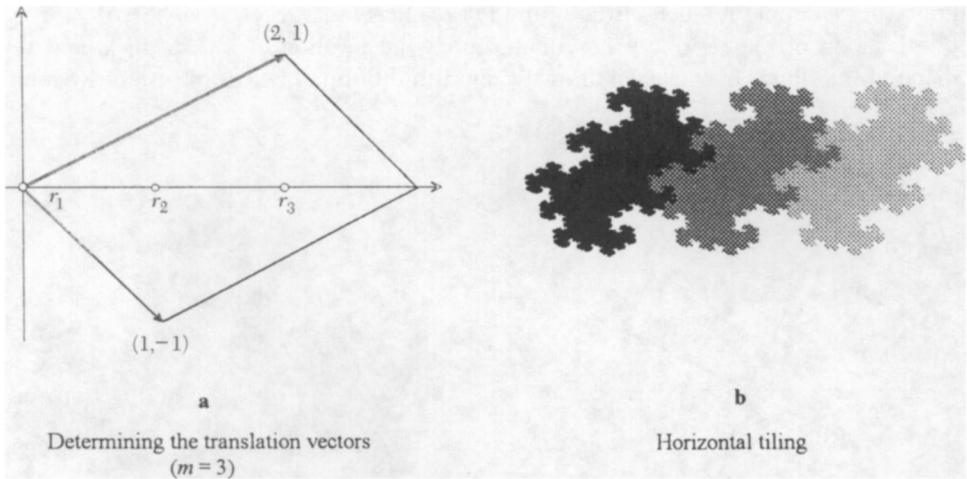


FIGURE 3

Generating the tilings

What characteristics of a matrix M and translation vectors $\mathbf{r}_j$ determine an iterative process that produces a desired tiling? We want the invertible integer matrix $M = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ to be an *expansive map*; i.e., all the eigenvalues of M have modulus larger than 1. A property of expansive maps is that, for some $n > 0$, M^{-n} is a *contraction mapping*; i.e., $|M^{-n}\mathbf{z}| < |\mathbf{z}|$. This ensures that iteration using the collection of functions $f_j = \mathbf{r}_j + M^{-1}\mathbf{z}$, $j = 1, \dots, m$, produces the attractor regardless of the choice of translation vectors $\{\mathbf{r}_j\}$ or of initial value $\mathbf{z}_0$. (The definition of convergence to an attractor, using an iterated function system, requires a knowledge of the Hausdorff metric. For more details, see chapter 2 of Barnsley [2].)

How are translation vectors chosen to produce tilings? For a matrix M as given above, $|\det(M)| = |ad - bc| = m$ is the area of the parallelogram P spanned by the vectors $\mathbf{v}_1 = \begin{bmatrix} a \\ c \end{bmatrix}$ and $\mathbf{v}_2 = \begin{bmatrix} b \\ d \end{bmatrix}$. Recall that the vectors in $\mathcal{E}$ are those with integer coordinates that lie in or on P , but not on the two edges that do not contain $\begin{bmatrix} 0 \\ 0 \end{bmatrix}$. We shall call these vectors the *principal residue vectors*. Let L denote the lattice of all points in the complex plane with integer coordinates. (These points are known as *Gaussian integers*.) Note that $\mathcal{E} \subseteq L$. For $j = 1, \dots, m$, define $L_j := \{\mathbf{r}_j + M\mathbf{x} : \mathbf{x} \in L\}$. The vectors $\{\mathbf{r}_j\}$ are said to form a *complete residue system* for M , because $L = \bigcup_{j=1}^m L_j$ and $L_j \cap L_k = \emptyset$ whenever $j \neq k$. (For more on complete residue systems, see Gilbert [4].)

The corner points of the parallelogram P are members of L and are linear combinations (with coefficients either 0 or 1) of the columns of M . All integer linear combinations of the columns of M form a subset G of L . In the plane, the subset G forms a grid of parallelograms, each congruent to P . Each parallelogram contains m points of L , just as P does. (We count the points in the congruent parallelograms with the same conventions as in P . See, for example, FIGURE 4a.) Each point $\mathbf{r}_j$ of L inside P is equivalent to one point $\mathbf{y}_j$ (and we write $\mathbf{r}_j \approx \mathbf{y}_j$) inside each of these congruent parallelograms. In general, as long as $\mathbf{y}_1 = \mathbf{r}_1 = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$ and $\mathbf{y}_j \approx \mathbf{r}_j$ for $j = 2, \dots, m$, then the collection $\{\mathbf{y}_j\}$ will also form a complete residue system for the matrix M . Following Example A, each of the four additional vectors was equivalent to vector $\mathbf{r}_2$. The location of the residue vectors determines the location of the fractiles, and the shape of the tilings may change dramatically with different choices of residue systems.

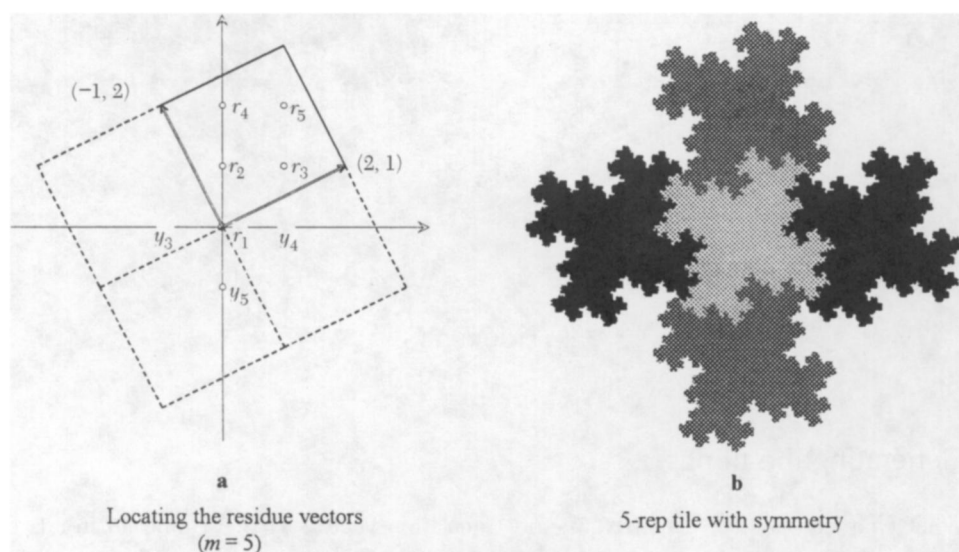


FIGURE 4

We can summarize these ideas as follows:

PROPOSITION. Suppose that M represents an expansive map, $\{\mathbf{y}_1, \dots, \mathbf{y}_m\}$ is a complete residue system for M , and $f_j(\mathbf{z}) = \mathbf{y}_j + M^{-1}\mathbf{z}$. Then the attractor set $A = \bigcup_{j=1}^m A_j$ is the union of m tiles A_j that have disjoint interiors and satisfy $A_j = f_j(A)$. Such tiles are called m -rep tiles. (This result is Theorem 1 in [1].)

We illustrate the construction of a tiling using m -rep tiles in the following example.

Example C. Let $M = \begin{bmatrix} 2 & -1 \\ 1 & 2 \end{bmatrix}$; then $m = 5$. As FIGURE 4a illustrates, the principal residue vectors are $\mathbf{r}_1 = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$, $\mathbf{r}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$, $\mathbf{r}_3 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$, $\mathbf{r}_4 = \begin{bmatrix} 0 \\ 2 \end{bmatrix}$, and $\mathbf{r}_5 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$. For a more symmetric tiling, we choose equivalent vectors for our residue system. FIGURE 4b is generated by setting $\mathbf{y}_1 = \mathbf{r}_1$, $\mathbf{y}_2 = \mathbf{r}_2$, $\mathbf{y}_3 = \begin{bmatrix} -1 \\ 0 \end{bmatrix} \approx \mathbf{r}_3$, $\mathbf{y}_4 = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \approx \mathbf{r}_4$, and $\mathbf{y}_5 = \begin{bmatrix} 0 \\ -1 \end{bmatrix} \approx \mathbf{r}_5$. Note that the vectors $\{\mathbf{y}_2, \mathbf{y}_3, \mathbf{y}_4, \mathbf{y}_5\}$, considered as complex numbers, are the fourth roots of unity and are symmetric about $\mathbf{r}_1$.

Constructing tiles with radial symmetry

In Example B, we can replace $\begin{bmatrix} 2 \\ 0 \end{bmatrix}$ with the equivalent vector $\begin{bmatrix} -1 \\ 0 \end{bmatrix}$ and use $\begin{bmatrix} 0 \\ 0 \end{bmatrix}$, $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$, and $\begin{bmatrix} -1 \\ 0 \end{bmatrix}$ as residue vectors. Observe that $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$ and $\begin{bmatrix} -1 \\ 0 \end{bmatrix}$ are symmetrically located about $\mathbf{r}_1 = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$. Can we always find residue vectors $\mathbf{r}_2, \dots, \mathbf{r}_m$ that are symmetrically located about $\mathbf{r}_1$? That is, can we construct a tiling that exhibits radial symmetry about $\mathbf{r}_1$? (See, for example, FIGURE 4b.) This turns out to be possible only when $m = 2, 3, 4, 5$, and 7 . (For an algebraic proof, see [8].) The cases $m = 2, 3$, and 5 were illustrated in previous examples. The cases $m = 4$ and $m = 7$ require more analysis.

For $m = 4$, the matrix $M = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ must represent an expansive map and have determinant 4. These conditions mean that $(a - \lambda)(d - \lambda) - bc = 0$ has roots λ with $|\lambda| > 1$ and that $ad - bc = 4$. Since $\lambda = \frac{1}{2}[(a + d) \pm \sqrt{(a + d)^2 - 16}]$, we have two cases: (i) If $(a + d)^2 - 16 < 0$, then λ is complex, $|a + d| < 4$ and $\lambda\bar{\lambda} = 4$. (ii) If λ is real, then $|a + d| \geq 4$, and $|\lambda| > 1$ implies that $|a + d| - \sqrt{(a + d)^2 - 16} > 2$, or $|a + d| < 5$. Thus $|a + d| = 4$. From (i) and (ii) we conclude that M must be chosen so that $|a + d| \leq 4$. In Example D, $a + d = 2$.

Example D. Let $M = \begin{bmatrix} 2 & -2 \\ 2 & 0 \end{bmatrix}$, with principal residue vectors as shown in FIGURE 5a. To achieve radial symmetry we want to generate a tiling using three vectors symmetrically located about $\mathbf{r}_1 = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$. The vectors $\mathbf{y}_1 = \mathbf{r}_1$, $\mathbf{y}_2 = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \approx \mathbf{r}_2$, $\mathbf{y}_3 = \begin{bmatrix} -1 \\ -1 \end{bmatrix} \approx \mathbf{r}_3$ and $\mathbf{y}_4 = \mathbf{r}_4 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$ form a complete residue system for M , but they are not symmetric about $\mathbf{r}_1$. We observe that the complex third roots of unity $\mathbf{v}_1 = \begin{bmatrix} -1/2 \\ \sqrt{3}/2 \end{bmatrix}$, $\mathbf{v}_2 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$, and $\mathbf{v}_3 = \begin{bmatrix} -1/2 \\ -\sqrt{3}/2 \end{bmatrix}$ are symmetrically located about $\mathbf{r}_1$, but that the vectors $\mathbf{v}_1$ and $\mathbf{v}_3$ are not Gaussian integers. If we apply the linear transformation represented

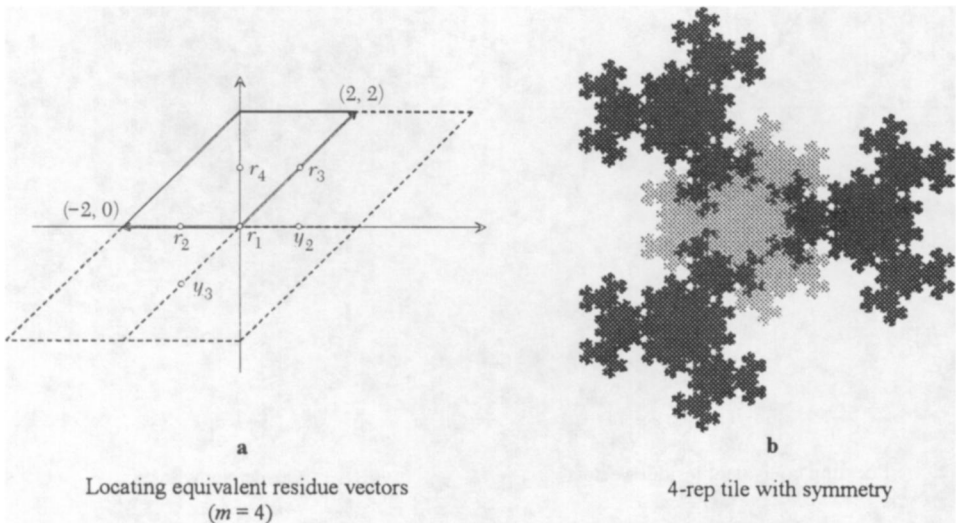


FIGURE 5

by $B = \begin{bmatrix} 1 & -1/2 \\ 0 & \sqrt{3}/2 \end{bmatrix}$, then we get $B\mathbf{y}_1 = \mathbf{y}_1$, $B\mathbf{y}_2 = \mathbf{v}_2$, $B\mathbf{y}_3 = \mathbf{v}_3$, and $B\mathbf{y}_4 = \mathbf{v}_1$. The iteration process can now be performed using the functions $f_j(\mathbf{z}) = B\mathbf{y}_j + h^{-1}(\mathbf{z})$, where $h = BMB^{-1}$ and $h^{-1} = \begin{bmatrix} 1/4 & \sqrt{3}/4 \\ -\sqrt{3}/4 & 1/4 \end{bmatrix}$. (Note that h^{-1} represents multiplication by the complex number $1/4 - i\sqrt{3}/4$.) The resulting tiling is shown in FIGURE 5b. The transformation B is the change of basis matrix that converts the lattice formed by $\mathcal{E}$ into one formed by $\mathbf{v}_1$ and $\mathbf{v}_2$. The reader is encouraged to try the same procedure for other matrices M with $|a + d| \leq 4$.

To see that, even after a change of basis, the transformations produce a tiling, set $A_j := f_j(A) = \mathbf{y}_j + M^{-1}(A)$. Then $A = \cup_{j=1}^m A_j$ and the A_j 's have disjoint interiors. Set $K_j := B(A_j)$ for each j and $K := B(A)$. Then $K_j = B(\mathbf{y}_j + M^{-1}(A)) = B\mathbf{y}_j + BM^{-1}B^{-1}B(A) = B\mathbf{y}_j + h^{-1}K$. It follows that $K = \cup_{j=1}^m K_j$ and that the K_j 's have disjoint interiors.

In the next example, we will construct a tiling with six unit vectors symmetrically located on the unit circle. We use the complex sixth roots of unity, $\mathbf{v}_j = \exp(i\pi j/3)$, $1 \leq j \leq 6$. In this case, $\det(M) = 7$; for reasons discussed prior to Example D, we must restrict $|a + d| \leq 7$.

Example E. Let $M = \begin{bmatrix} 1 & -2 \\ 2 & 3 \end{bmatrix}$, with principal residue vectors $\{\mathbf{r}_j\}$ as shown in FIGURE 6a. Note that we can choose the residue vectors $\mathbf{y}_1 = \mathbf{r}_1$, $\mathbf{y}_2 = \mathbf{r}_2$, $\mathbf{y}_3 = \begin{bmatrix} -1 \\ 1 \end{bmatrix} \approx \mathbf{r}_3$, $\mathbf{y}_4 = \begin{bmatrix} -1 \\ 0 \end{bmatrix} \approx \mathbf{r}_4$, $\mathbf{y}_5 = \begin{bmatrix} 0 \\ -1 \end{bmatrix} \approx \mathbf{r}_5$, $\mathbf{y}_6 = \begin{bmatrix} 1 \\ -1 \end{bmatrix} \approx \mathbf{r}_6$ and $\mathbf{y}_7 = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \approx \mathbf{r}_7$. We set $B = \begin{bmatrix} 1 & 1/2 \\ 0 & \sqrt{3}/2 \end{bmatrix}$ and note that $B\mathbf{y}_{j+1} = \mathbf{v}_j$, $1 \leq j \leq 6$. We now iterate using the functions

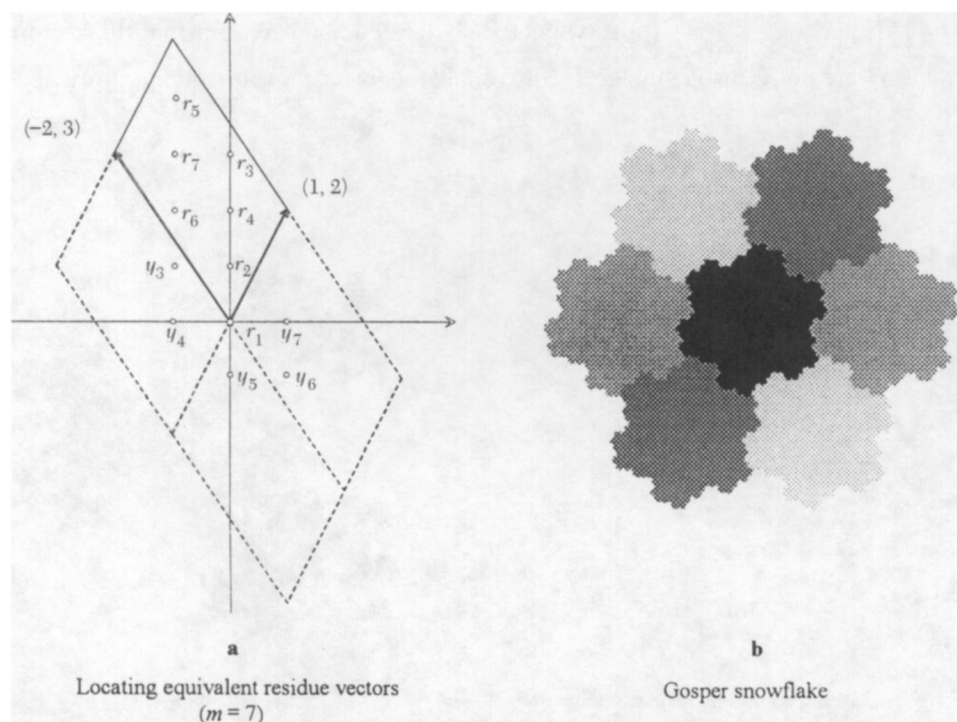


FIGURE 6

$f_j(\mathbf{z}) = B\mathbf{y}_j + h^{-1}(\mathbf{z})$, $j = 1, \dots, 7$, where $h^{-1} = B^{-1}M^{-1}B = \begin{bmatrix} 2/7 & \sqrt{3}/7 \\ -\sqrt{3}/7 & 2/7 \end{bmatrix}$. Again

we note that the matrix h^{-1} represents multiplication by the complex number $2/7 - i\sqrt{3}/7$. The resulting tiling, called the Gosper snowflake, is shown in FIGURE 6b. The Gosper snowflake changes the regular hexagon “just enough” to allow a subdivision into seven similar fractiles.

Similarity maps

A similarity map g satisfies $|g(\mathbf{x}) - g(\mathbf{y})| = r|\mathbf{x} - \mathbf{y}|$, for $r > 0$ and all $\mathbf{x}, \mathbf{y}$ in the plane. Geometrically, a similarity map is a composition of any collection of the four simple mappings: scaling by a positive factor r , rotation about the origin, translation, and reflection. If each mapping in our collection $\{f_j\}$ is a similarity map with $0 < r < 1$, then the resulting tiling (attractor) A is *self-similar*. That is, A is the union of m smaller copies of itself. In this case, the attractor has the same shape as each of its m -rep tiles. This phenomenon appears in FIGURES 2b, 4b, 5b and 6b. Each of these rep tiles arises from a special sort of linear map: multiplication by a complex number.

Multiplication by a complex number

As we have seen, each complex number $q = \alpha + i\beta$ corresponds to the matrix

$$h = \begin{bmatrix} \alpha & -\beta \\ \beta & \alpha \end{bmatrix};$$

then the matrix operations correspond to ordinary arithmetic operations on complex numbers. If $|q| > 1$, the map h is expansive, since the eigenvalues of h have modulus $\alpha^2 + \beta^2 = |q|^2$. Also, h is a similarity map, since if $\mathbf{z} = (x_1, x_2)$, then

$$|h(x_1, x_2)|^2 = (\alpha x_1 - \beta x_2)^2 + (\beta x_1 + \alpha x_2)^2 = (\alpha^2 + \beta^2)(x_1^2 + x_2^2) = |q|^2 |x|^2.$$

FIGURE 2b is derived from the complex number $1 - i$ and FIGURE 4b from $2 + i$.

If α and β are not integers, but $|q| > 1$, a change of basis can still produce a similarity map h from an integer matrix M . For example, define the change of basis matrix

$$B := \begin{bmatrix} 1 & \alpha \\ 0 & -\beta \end{bmatrix};$$

set

$$M = \begin{bmatrix} 2\alpha & \alpha^2 + \beta^2 \\ -1 & 0 \end{bmatrix};$$

and let $h = BMB^{-1}$. If α and β are chosen so that M is an integer matrix, all earlier methods can be applied. We illustrate the process in the next example.

Example F. Let $q = \frac{1}{2} + i\sqrt{15}/2$. With $B = \begin{bmatrix} 1 & 1/2 \\ 0 & -\sqrt{15}/2 \end{bmatrix}$ we find that $M = \begin{bmatrix} 1 & 4 \\ -1 & 0 \end{bmatrix}$, and $\mathbf{r}_1 = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$, $\mathbf{r}_2 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$, $\mathbf{r}_3 = \begin{bmatrix} 2 \\ 0 \end{bmatrix}$, and $\mathbf{r}_4 = \begin{bmatrix} 3 \\ 0 \end{bmatrix}$. A tiling can be derived

using the functions $f_j = B\mathbf{y}_j + h^{-1}(\mathbf{z})$, where $\mathbf{y}_1 = \begin{bmatrix} 1 \\ -1 \end{bmatrix} \approx \mathbf{r}_1$, $\mathbf{y}_2 = \mathbf{r}_2$, $\mathbf{y}_3 = \begin{bmatrix} -1 \\ -1 \end{bmatrix} \approx \mathbf{r}_3$, $\mathbf{y}_4 = \begin{bmatrix} -1 \\ 0 \end{bmatrix} \approx \mathbf{r}_4$ and $h = BMB^{-1}$.

Developing similarity maps

In Examples D and E, a change of basis applied to M produced similarity mappings and attractive tilings. This process works—i.e., we can choose a matrix B such that $h = BMB^{-1}$ is a similarity mapping—if M either has (i) two real eigenvalues with equal modulus and independent eigenvectors or (ii) a pair of complex conjugate eigenvalues.

In case (i), let λ_1 and λ_2 ($\lambda_1 = \pm\lambda_2$) be real eigenvalues for M with corresponding eigenvectors $\mathbf{v}_1$ and $\mathbf{v}_2$. Let $B^{-1} = [\mathbf{v}_1, \mathbf{v}_2]$ be the matrix with column vectors $\mathbf{v}_1$ and $\mathbf{v}_2$. Then

$$MB^{-1} = M[\mathbf{v}_1, \mathbf{v}_2] = [\lambda_1\mathbf{v}_1, \lambda_2\mathbf{v}_2] = B^{-1} \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix}$$

so that $h := BMB^{-1} = \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix}$ is a similarity map. Then h can be used to generate the tiling with functions of the form $f_j = B\mathbf{y}_j + h^{-1}(\mathbf{z})$, where $\{\mathbf{y}_j\}$ are residue vectors for M . (Note that h depends only on the choice of *eigenvalues* while the translation vectors, and the tiling, vary with the choice of *eigenvectors*.) Example G illustrates this case.

Example G. The matrix $M = \begin{bmatrix} 2 & 2 \\ 1 & -2 \end{bmatrix}$ has determinant -6 and eigenvalues $\lambda = \pm\sqrt{6}$. Associated eigenvectors of M are $\mathbf{v}_1 = \begin{bmatrix} 1 \\ (\sqrt{6}-2)/2 \end{bmatrix}$ and $\mathbf{v}_2 = \begin{bmatrix} 1 \\ (\sqrt{6}+2)/2 \end{bmatrix}$ so $B^{-1} = \begin{bmatrix} 1 & 1 \\ (\sqrt{6}-2)/2 & (\sqrt{6}+2)/2 \end{bmatrix}$ and the similarity map is $h = \begin{bmatrix} \sqrt{6} & 0 \\ 0 & -\sqrt{6} \end{bmatrix}$. FIGURE 7 shows the tiling generated using the functions $f_j(\mathbf{z}) = B\mathbf{r}_j + h^{-1}(\mathbf{z})$ with principal residue vectors

$$\mathbf{r}_1 = \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \mathbf{r}_2 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \mathbf{r}_3 = \begin{bmatrix} 2 \\ 0 \end{bmatrix}, \mathbf{r}_4 = \begin{bmatrix} 2 \\ -1 \end{bmatrix}, \mathbf{r}_5 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}, \mathbf{r}_6 = \begin{bmatrix} 3 \\ -1 \end{bmatrix}.$$



FIGURE 7

Similarity tiling using a change of basis ($m = 6$)

In case (ii), suppose that $M = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ has complex conjugate eigenvalues $\lambda = \alpha + i\beta$

and $\bar{\lambda} = \alpha - i\beta$, $\beta \neq 0$. Let $\mathbf{v} = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} v_{11} + iv_{12} \\ v_{21} + iv_{22} \end{bmatrix}$ be an eigenvector associated with λ .

Then $B^{-1} = \begin{bmatrix} v_{11} & v_{12} \\ v_{21} & v_{22} \end{bmatrix}$ is the inverse of an appropriate change of basis matrix for obtaining a similarity transformation h . To see this, keep in mind that the real and imaginary parts of $M\mathbf{v}$ and $\lambda\mathbf{v}$ must be the same, and use the same strategy as in case (i) to obtain $MB^{-1} = B^{-1} \begin{bmatrix} \alpha & \beta \\ -\beta & \alpha \end{bmatrix}$. Therefore, $h = \begin{bmatrix} \alpha & \beta \\ -\beta & \alpha \end{bmatrix}$, which represents multiplication by the complex number λ . Since B^{-1} is defined using an eigenvector of M , the matrix B can have many forms. For example, if we choose $v_1 = 1 + iv_{12}$ and $v_2 = 0 + iv_{22}$, we find that

$$B^{-1} = \begin{bmatrix} 1 & \frac{1}{\beta}(\alpha - a) \\ 0 & \frac{-c}{\beta} \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 1 & \frac{(\alpha - a)}{c} \\ 0 & \frac{-\beta}{c} \end{bmatrix}$$

as illustrated in Example H. If, as in Example I, we choose $v_1 = 1 + i0$, we have

$$B^{-1} = \begin{bmatrix} 1 & 0 \\ \frac{\alpha - a}{b} & \frac{-\beta}{b} \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 1 & 0 \\ \frac{\alpha - a}{\beta} & \frac{-b}{\beta} \end{bmatrix}.$$

In both cases, the transformation $h = BMB^{-1}$ is a similarity mapping.

Example H. Let $M = \begin{bmatrix} 1 & -1 \\ 1 & 2 \end{bmatrix}$; then M has determinant 3 and eigenvalues $\frac{3}{2} \pm i\sqrt{3}/2$. With $B = \begin{bmatrix} 1 & \frac{1}{2} \\ 0 & -\sqrt{3}/2 \end{bmatrix}$, the similarity mapping $h = BMB^{-1}$ is given by the matrix $\begin{bmatrix} \frac{3}{2} & \sqrt{3}/2 \\ -\sqrt{3}/2 & \frac{3}{2} \end{bmatrix}$. If we choose the residue vectors $\mathbf{y}_1 = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$, $\mathbf{y}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$, and $\mathbf{y}_3 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ and iterate using the functions $f_j = B\mathbf{y}_j + h^{-1}(\mathbf{z})$, $j = 1, 2, 3$, we obtain the so-called terdragon shown in FIGURE 8. (For comparison, see again the 3-rep tile of Example B, and note the subtle changes in the matrices.)

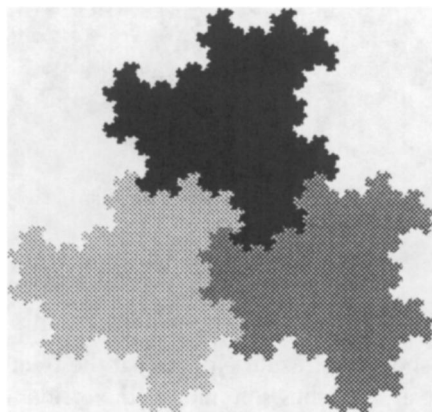


FIGURE 8
The Terdragon (a 3-rep tile)

Example I. Let $M = \begin{bmatrix} 1 & -1 \\ 2 & 3 \end{bmatrix}$; then $\det(M) = 5$ and M has eigenvalues $\lambda = 2 \pm i$. The change of basis matrix $B = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$ yields the similarity transformation $h = BMB^{-1} = \begin{bmatrix} 2 & -1 \\ 1 & 2 \end{bmatrix}$. With the residue vectors

$$y_1 = \begin{bmatrix} 0 \\ 0 \end{bmatrix}, y_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, y_3 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}, y_4 = \begin{bmatrix} -1 \\ 1 \end{bmatrix}, y_5 = \begin{bmatrix} 0 \\ -1 \end{bmatrix},$$

we obtain the 5-rep tile shown in FIGURE 4b.

Variations

Once one begins to generate tilings as above, ideas for modifying the figures abound. We demonstrate a few variations below; readers are encouraged to experiment further.

Example J. What happens if one of the functions is removed from the iteration process? Using the similarity transformation from Example E, we generated the wreath in FIGURE 9 by omitting the function with residue vector r_1 . (Compare FIGURE 6b.)

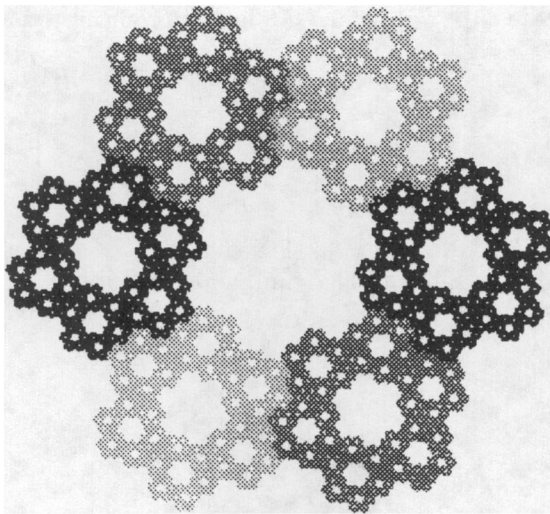


FIGURE 9
Wreath (modified snowflake)

Example K. Let $M = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$ and use the residue vectors y_1, v_1, v_2 , and v_3 from Example D. FIGURE 10a shows the result. Notice that the fractiles appear to overlap. In fact, they do not. FIGURE 10b shows a modified version of the generated tiling, omitting the piece A_1 associated with the y_1 residue vector. This shows that the fractiles $\{A_j\}$ are not simply connected.

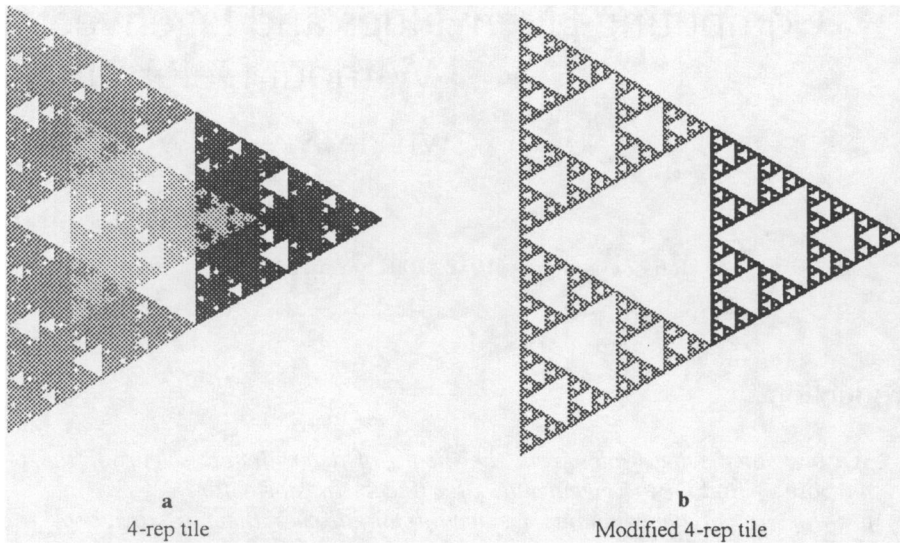


FIGURE 10

Note. Many computer resources are available for generating fractals. The Random Iteration Algorithm, presented by Barnsley in [2], can be used to generate pictures like those in this paper. *Fractal Attraction* [11] is another useful tool for investigating these ideas. FRACTINT, used to generate the fractals in this article, is freeware, available from <http://spanky.triumf.ca/www/fractint/getting.html>. Generating fractals in color presents even more dramatic pictures.

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